

Global-Toy Company Sales Analysis

Insights for Product and Seasonal Strategy (2003-2005)

Executive Summary:

This report analyzes Global-Toy company sales data from 2003 to 2005 to identify trends, top-performing products, and key geographic markers. The analysis highlights significant seasonality and strong product and regional concentration, which should inform immediate strategic decisions.

Table 1: Short Summary:

Key Finding	Conclusion	Actionable Recommendation
Sales show Extreme Seasonality , with Q4 contributing the highest revenue consistently(Peaking at \$1.66M in Q4 2004).	The company is heavily reliant on the holiday season for revenue and must optimize preparation.	Prioritize Q4 Inventory & Marketing: Implement an aggressive Q4 inventory and promotional strategy to maximize holiday season revenue.
Classic Cars are the dominant product line, generating \$2.97M (significantly more than other categories).	Classic Cars are the core revenue driver and must be protected and maximized.	Focus Product Strategy: Allocate the majority of marketing budget and product development resources to the Classic Cars line.
The USA is the largest market, contributing \$2.99M in sales, followed by Spain \$1.02M and France \$919K .	Sales concentration in the USA poses both an opportunity and a risk if the market falters.	Prioritize Us Retention & Secondary Market Growth: Focus retention efforts on the USA while actively developing Spain and France as strong secondary markers.

Introduction:

This Section aims to provide richer context for the business problem and the value proposition of your analysis.

1. Business Objective:

The toy market is highly seasonal and competitive. The company is currently making decisions based on fragmented, raw transaction logs, leading to **sub-optimal inventory levels** and **misaligned marketing spend**. Specifically, management needs answers to three core questions:

- a) Are sales growth rates consistent across all product lines?
- b) Which markets should receive priority investment?
- c) How pronounced is our reliance on holiday sales?

2. Goal:

To transition the business from reactive reporting to **proactive, data-driven decision-making**. This project delivers a **single source of truth**—an interactive Power BI dashboard—to enable continuous monitoring of key sales performance indicators (KPIs), forecast future demand, and provide specific, actionable insights into product and geographic strategy.

3. Scope:

The analysis covers all available transaction data from **January 2003 through June 2005 (Q2)**. It focuses solely on reported sales data (revenue, quantity, and status) and excludes external factors like competitor performance or macroeconomic trends. All findings and recommendations are limited to the provided dataset.

Methodology:

1. Data source and Pipeline:

The raw dataset was a large, transaction-level CSV file sourced from Kaggle. The data was ingested and processed via a script-based pipeline (**R/Python**) to ensure reproducibility and consistency before being loaded into Power BI. This structured pipeline is key for future automated report refreshes.

2. Data Cleaning:

The cleaning process was non-trivial, designed to eliminate the risk of corrupted insights:

1. **Duplicate Handling:** A composite key (ORDERNUMBER + ORDERLINENUMBER) was necessary to ensure uniqueness, as duplicate rows would inflate sales figures.
2. **Validation of Numeric Fields:** Critical sales fields were checked for non-positive values, which were flagged as transactional errors.
3. **Sales Integrity Check:** Recalculating and confirming that $\text{SALES} = \text{QUANTITYORDERED} \times \text{PRICEEACH}$ served as a **critical audit step** to guarantee revenue accuracy.
4. **Date Validation:** Standardizing dates ensured accurate time-series analysis for seasonality.

3. Analytical & Visualization Techniques:

Business Intelligence (BI) Environment: Power BI was chosen for its strong integration capabilities and user-friendly, interactive dashboard features. **DAX Measures:** Custom Data Analysis Expressions (DAX) were created to derive essential business metrics: * **Average Order Value (AOV):** Total Sales / Total Number of Orders. * **Sales Contribution:** Calculated as a percentage of total revenue for each country and product line. **Visualization:** Used best practices to ensure clarity: Line charts for trends (time-series), Bar charts for comparison (product performance), and Map visuals for geographic impact

4. Outcome:

The rigorous methodology resulted in a **validated, analysis-ready dataset** (sales_data_cleaned.csv) and a robust dashboard, providing a high degree of confidence in the final insights and recommendations.

Findings:

1. Sales Trend Over Time (Seasonality)

Sales show a clear and significant seasonal pattern, with revenue spiking sharply in the fourth quarter of every year analyzed. The highest sales overall occurred in **Q4 2004 with \$1.66M**.

Year	Q1	Q2	Q3	Q4
2003	\$372k	\$473k	\$527k	\$1.53M
2004	\$693k	\$639k	\$921k	\$1.66M
2005	\$889k	\$590k	N/A	N/A

2. Top-Selling Product Lines

The **Classic Cars** line is the clear revenue leader, generating almost double the sales of the next closest product line, **Vintage Cars**.

Product Line	Total Sales
Classic Cars	\$2.97M
Vintage Cars	\$1.64M
Motorcycles	\$971k
Trucks and Buses	\$947k
Planes	\$878k

3. Geographical Insights

The company's sales are primarily driven by the **USA**, which accounts for nearly three times the revenue of the next country, **Spain**.

Country	Total Sales
USA	\$2.99M
Spain	\$1.02M
France	\$919k
Australia	\$522k
UK	\$413k

4. Sales Status

Order fulfillment status was tracked, with the majority of orders marked as 'Shipped.' The analysis also provided tracking on canceled and on-hold orders via the dashboard visualization.

Conclusions

The project successfully achieved its objective by quantifying sales performance across time, product, and region.

- **Seasonality dictates strategy:** The Q4 sales spike confirms strong seasonality, making it the most critical period for operational and marketing focus.
- **Product focus is narrow:** The overwhelming dominance of Classic Cars indicates a reliance on a single product category, which presents an area for strategic focus but also a vulnerability.
- **Regional strategy is clear:** The USA is the most valuable market, followed by a strong second-tier cluster of Spain and France, which should be the target of secondary growth initiatives.

Recommendations

Based on the findings, the following actions are recommended for the business stakeholders:

- ✓ **Operational Planning:** Implement detailed inventory and fulfillment planning to support the predictable, extreme demand spikes in Q4.
- ✓ **Marketing Allocation:** Re-allocate marketing spend to prioritize the **Classic Cars** line, ensuring maximum visibility and promotion for this core product.
- ✓ **Market Development:** Invest resources in **Spain and France** to convert them from secondary to primary high-growth markets, diversifying risk away from the heavy reliance on the USA.
- ✓ **Dashboard Utilization:** Use the **interactive Power BI dashboard** (in dashboard folder) for weekly monitoring of key performance indicators (KPIs) like Total Sales and Average Order Value (AOV).

Appendix

- ❖ Code scripts used for data cleaning (R and Python) are available in the Code folder.
- ❖ Detailed list of all data cleaning steps performed, including logic for the Sales Integrity Check ($\$SALES = QUANTITYORDERED \times PRICEEACH\$$) and column selection.
- ❖ Screenshots of the interactive Power BI Dashboard are provided in `Dashboard_page-1&2.PNG`.